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**MR-21**

**Code: B415CS5S2**

# MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

## Computer Organization and Architecture

*(Computer Science and Engineering)*

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

***Part-A (10 x 2 = 20 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 1. a) | What is the difference between computer design and computer architecture? | 2 | 1 | 1 | 1 |
| 1. b) | Define instruction code and instruction cycle. | 2 | 2 | 1 | 2 |
| 1. c) | What is a control memory? | 2 | 2 | 2 | 2 |
| 1. d) | Define addressing mode with an example. | 2 | 1 | 2 | 1 |
| 1. e) | Define 1's and 2's complements. | 2 | 2 | 3 | 3 |
| 1. f) | What is the difference between fixed-point and floating-point representation? | 2 | 3 | 3 | 3 |
| 1. g) | What is DMA? Why is it used? | 2 | 2 | 4 | 2 |
| 1. h) | What are different types of mapping in cache memory? | 2 | 2 | 4 | 3 |
| 1. i) | What is parallel processing? | 2 | 2 | 5 | 1 |
| 1. j) | Mention any two characteristics of RISC processor. | 2 | 2 | 5 | 2 |

***Part-B (5 x 10 = 50 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 2. a) | Explain the basic computer organization with block diagram. | 5 | 2 | 1 | 1 |
| 2. b) | Explain the various types of micro-operations. | 5 | 3 | 1 | 2 |
| | **OR** | | | | |
| 2. c) | Explain the function of arithmetic logic shift unit with neat diagram. | 5 | 3 | 1 | 2 |
| 2. d) | Write short notes on bus and memory transfer operations. | 5 | 2 | 1 | 3 |
| 3. a) | Explain the design of a microprogrammed control unit. | 5 | 2 | 2 | 3 |
| 3. b) | Discuss various types of instruction formats. | 5 | 2 | 2 | 2 |
| | **OR** | | | | |
| 3. c) | Explain in detail about data transfer and manipulation instructions. | 5 | 2 | 2 | 2 |
| 3. d) | Explain the role of a sequencer in a control unit. | 5 | 3 | 2 | 3 |
| 4. a) | Explain signed and unsigned number representation. | 5 | 3 | 3 | 3 |
| 4. b) | Describe the steps involved in floating-point addition and subtraction. | 5 | 3 | 3 | 3 |
| | **OR** | | | | |
| 4. c) | Explain multiplication and division algorithms with examples. | 5 | 2 | 3 | 3 |
| 4. d) | Discuss decimal arithmetic operations and hardware implementation. | 5 | 2 | 3 | 2 |
| 5. a) | Explain different modes of data transfer in I/O systems. | 5 | 3 | 4 | 3 |
| 5. b) | Explain cache memory organization and mapping techniques. | 5 | 3 | 4 | 3 |
| | **OR** | | | | |
| 5. c) | Draw and explain the working of a DMA controller. | 5 | 3 | 4 | 1 |
| 5. d) | Explain priority interrupt and its control mechanism. | 5 | 2 | 4 | 4 |
| 6. a) | Compare and contrast RISC and CISC architectures. | 5 | 4 | 5 | 4 |
| 6. b) | Explain instruction pipeline and its advantages. | 5 | 4 | 5 | 2 |
| | **OR** | | | | |
| 6. c) | Explain characteristics of multiprocessor systems. | 5 | 4 | 5 | 2 |
| 6. d) | Explain vector processing and array processors. | 5 | 5 | 5 | 3 |

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome